

Propagation Structure as a Signal for Misinformation Detection with Graph Neural Networks and Edgeless Baselines

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Abstract. Propagation-based detectors classify a story from the shape of the cascade that carries it, and they report strong benchmark figures. This article measures what those figures are worth. On UPFD, a bidirectional graph convolutional network reaches macro-F1 0.832 on PolitiFact and 0.920 on GossipCop. Handed the same node features with every edge deleted, a logistic regression reaches **0.940** on the second, and one integer, the number of accounts, reaches 0.723. A paired bootstrap finds it ahead on one of four configurations, +0.103, behind on another, -0.021, and indistinguishable on the two smallest. The configuration it loses is one where the benchmark hands it the news article, and masking that article reverses the outcome on GossipCop while costing accuracy on PolitiFact, so **what the article embedding is worth changes sign between corpora**, by more than the architectural differences this literature reports. **Macro-F1 is already at its level from a fifth of a cascade observed.** An influence ranking over reach, PageRank and k-core answers negatively on real data: against a 14.5% chance baseline set by unequal cascade sizes, the hundred most influential accounts sit at 13.4%. Carried onto 400 cascades collected from Bluesky the detector collapses, training from scratch beats fine-tuning at every data budget, and a one-parameter rule on cascade size outscores it, 0.902 against 0.768. What a propagation figure is worth depends on a baseline that reads no edge and on a feature supplied at the root, neither of which the compared studies report.

Keywords: misinformation detection, graph neural networks, propagation cascades, early detection, influence, cross-platform transfer, confounds.

1 Introduction

False stories travel farther, faster and deeper than true ones, and the effect is driven by people rather than by automation [1]. That finding is the empirical foundation of an entire strand of detection work: if falsehood diffuses differently, then the diffusion itself is a signal, and a classifier can read it without understanding the claim at all. The strand runs from recurrent models over engagement time series to graph neural networks over reshare trees [2, 3, 4], and it reports figures well above what content-only classifiers achieve on comparable tasks [5, 6].

This article reports such a system and then spends most of its length establishing what its figures are worth.

The detector performs as the literature suggests, and performs from a fifth of the cascade. Establishing that the graph rather than its node features is responsible is harder than it appears. Under a paired test, a linear model over the same node features with every edge deleted beats

the graph model on one of the four benchmark configurations and is indistinguishable from it on two more. The configuration it loses is one where the benchmark hands the model the news article at the cascade root, and removing that article reverses the outcome. What the article is worth to a propagation detector turns out to depend on the corpus and to change sign, costing up to 0.044 on one and earning up to 0.021 on the other.

The detector does not survive a change of platform, and the repair the transfer-learning literature would prescribe does not close the gap, though the size of the shortfall depends on giving both arms the same optimisation budget. When the collected labels are tested against simpler predictors, most of the apparent structural signal is accounted for by the audience size of the sampled outlets.

Four questions organise the work, each stated below with the objective that answers it.

- **RQ1.** Does propagation structure separate misleading from reliable content, and how early in the life of a cascade?

RO1. Benchmark three graph architectures on two corpora against baselines that delete every edge, test each margin with a paired bootstrap, and repeat the measurement on cascades truncated to five levels of observation.

- **RQ2.** Which account features identify the participants who drive the spread, and do those participants carry misinformation?

RO2. Score every account with a composite of reach, PageRank and k-core, then evaluate the resulting ranking against annotated veracity rather than against a proxy for spreading power.

- **RQ3.** What happens when a detector trained on one platform’s cascades meets another platform’s, and what does recovery cost?

RO3. Apply the benchmark-trained detector unchanged to cascades collected from a second platform, then compare fine-tuning against training from scratch at three data budgets under one optimisation budget.

- **RQ4.** What does an external content signal add to a structural detector?

RO4. Write a cross-fitted content credibility score onto the cascade root and ablate it against a random control, in every benchmark configuration.

RQ4 is answered in Section 5, from the network side of the interface. The content classifier that supplies that signal is reported in a separate submission by the same authors, drawn from the same project and disclosed here in full.

This article makes five contributions.

- Every margin we claim for the propagation structure is tested against a model that deletes every edge, under a paired bootstrap on the difference. The graph model separates from it on one of four benchmark configurations (Section 4.2).
- Ablating the news article UPFD places at the cascade root (Section 4.2) shows its contribution reversing in sign between the two subsets of one benchmark, by more than the architectural differences this literature reports (Section 4.3).
- A detection curve against the share of the cascade observed, on 3,826 test cascades, with the class shapes that accompany it (Section 4.3).
- An influence ranking evaluated against annotated veracity with a baseline corrected for unequal cascade sizes, answering negatively (Section 4.5).

- A collection protocol for a platform outside the benchmark ecosystem (Section 3.5), a measurement of transfer onto it (Section 4.7), and a control against our own best result there, following Pelrine et al. [7] and Wu and Hooi [8], which reduces the claim we can make from it (Section 4.9).

2 Related work

2.1 From sequences to graphs

Early rumour detection modelled engagement as a time series and applied recurrent networks to it [2]. Recursive models over the conversation tree followed, exploiting the structure the sequence discards [3]. The generalisation of convolution to graphs [9] and its attention-weighted variant [10] made cascade classification a standard graph-level task.

Bi-GCN [4] is the model we adopt. It builds two views of the same cascade, one following the direction of propagation from the source outward and one following the reversed edges, runs a graph convolution on each, pools both and concatenates the representations before classifying. The stated intuition is that propagation and dispersion carry different information: how a story spread, and how its periphery aggregates back toward the source.

UPFD [5] contributed both a method and the dataset used here, pairing each cascade with node features derived from the posting history of the participating accounts. Recent surveys place hybrid content-plus-structure models at the centre of the field [6, 11]. We keep the two signals separable rather than fused, for reasons developed in Section 5.

2.2 Influence and superspreaders

Influence maximisation established the algorithmic framing of who matters in a diffusion process [12]. Kitsak et al. showed that k-core position predicts spreading power better than degree, which is the result our composite score is built on [13]. Subsequent work established that a small number of accounts drive a disproportionate share of low-credibility spread [14, 15], and also that structural position alone does not settle who those accounts are. Our influence result is a direct, negative test of the second point on annotated data.

2.3 Benchmark shortcuts and trivial baselines

A line of work predating ours asks not how well a detector scores on these resources but what it is scoring on. Pelrine et al. reached the state of the art on Twitter16 by classifying on the first three digits of a tweet identifier, which encode its date [7]. Wu and Hooi showed that holding events out of the training split drops leading rumour detectors by more than 40 %, to the level of a trivial classifier, and proposed event-separated evaluation as the remedy [8]. The phenomenon is now studied under the name shortcut learning [16]. Section 4.9 applies their protocol, with source domains in the role of events, and reports what it costs us.

2.4 Platform provenance of benchmarks

UPFD is built on FakeNewsNet [17], and PHEME [18] serves conversation-level rumour work. Surveys of the area [19, 20, 21, 22] treat these as the standard resources. All of them are Twitter-era. Evaluation practice built on single splits of such resources has been criticised [23], and that criticism motivated both the domain-grouped conditions we report and the decision to collect from a platform outside the ecosystem. Bluesky is the practical choice, its protocol exposing documented read endpoints, and it has been characterised as a research object in its own right [24].

3 Method

3.1 Benchmark data and node features

UPFD [5] supplies one graph per news story: a root representing the article and one node per account that reshared it. We use both subsets, PolitiFact with 62 training and 221 test graphs and GossipCop with 1,092 and 3,826. Each subset also carries a validation split, 31 graphs on PolitiFact and 546 on GossipCop, which brings PolitiFact to 314 graphs in total. Model selection is fixed in advance rather than tuned, so the validation graphs enter only where a count over the whole subset is reported.

Two node feature sets are available and the distinction runs through every result. The *profile* set is ten hand-made counters per account: a verification flag, a geolocation flag, followers, friends, listed count, favourites, statuses, account age, screen-name length and description length. The *bert* set is 768-dimensional and 77 times wider, requiring a GPU, and it is not uniform across the cascade: each account node carries an embedding of that account’s posting history, while the root node carries an embedding of the news article itself. A model given the *bert* features is therefore reading text, precompiled and at one remove, and any structural claim made from those figures has to be read with that in mind. Section 4.2 asks what these features give without any graph at all, and Section 4.3 what the article at the root is worth once the graph is there. Reporting both feature sets is what allows the contribution of feature richness to be separated from the contribution of architecture.

3.2 Architectures and protocol

GCN [9] and GAT [10] are the baselines and Bi-GCN [4] the main model. All three are trained under one protocol so that differences are architectural: Adam, learning rate 0.01, 60 epochs, hidden dimension 64, mean pooling, fixed seeds. Figure 1 shows the main model. Implementation uses PyTorch [25] and PyTorch Geometric [26].

Two conventions govern every number reported below. Experiments are separate fits: a configuration that appears in more than one table was trained once per experiment under this protocol rather than trained once and reused, so the same cell can differ between tables by up to 0.004 where the model reads the same input. Where the input itself changes width, as in the ablation of Table 10, the difference is larger and that table states it. Differences quoted in the text are taken on the unrounded means and can therefore fall 0.001 away from the subtraction of two rounded cells.

Figure 1 sets out the architecture. A cascade is read twice, once along the direction it spread and once against it, and the two readings are pooled and concatenated before classification. The architecture never reads a sentence, but what reaches it depends on the feature set: under *profile* the nodes are account counters and no text enters, while under *bert* they are text embeddings and it does.

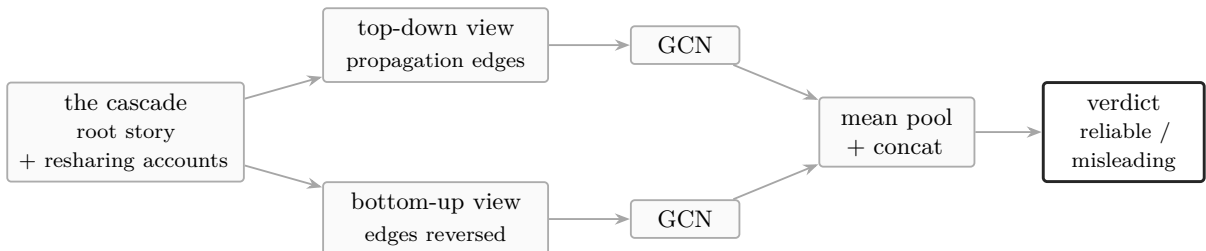


Figure 1. The detector, which reads each cascade in both directions before pooling.

3.3 Early detection by truncation

A detector that needs a complete cascade is of no operational use, since by the time the cascade is complete the story has already spread. We adopt the truncation protocol of Bi-GCN’s own early-detection experiment [4]: the model is trained on complete cascades and evaluated on truncated ones, keeping the first 20, 40, 60, 80 and 100 percent of nodes. UPFD carries no timestamps, so breadth-first order from the root stands in for order of arrival. This is an approximation and we name it as one.

3.4 Influence scoring

Each account receives a composite score, $0.5 \cdot \text{reach} + 0.3 \cdot \text{PageRank} + 0.2 \cdot \text{k-core}$, combining a direct behavioural measure with two structural ones, following the strands identified in the related work above. The weights are a stated design choice and are not fitted, which invites the question of how much the ranking owes to them. We answered it by recomputing the score on the collected network under seven alternative weightings, spanning from reach-dominant (0.7, 0.2, 0.1) through an equal split to core-dominant (0.2, 0.3, 0.5). Against the reference ranking, Spearman correlation never falls below **0.998**, and the hundred most influential accounts share between **93 and 99** of their members. The ranking is therefore a property of the three measures rather than of the particular weights placed on them. We report the ranking’s evaluation against annotated veracity rather than against a proxy for spreading power, because the question of interest is not who diffuses but who diffuses misinformation.

3.5 Collecting and labelling a sample outside the benchmark

The route that failed. Our first collection retrieved posts by keyword on misinformation-adjacent topics, 681 cascades over 109,491 accounts. The 140 that passed the check-worthiness gate were annotated by hand, together with 25 of those it rejected as a check on the gate itself, and that annotation yielded **87 reliable claims against 6 misleading ones**, the remaining 72 records stating nothing a verdict could be about. On this platform the accounts engaging with such topics are predominantly those refuting them. Six positives cannot support a metric, and the failure is itself a finding about the platform.

The route that worked. Posts are retrieved by the domain they link to, and the label is read from the published factual rating of that outlet, using the Iffy Index built on Media Bias/Fact Check. This yields **200 cascades per class** over 34 domains. Three filters are required: only the two lowest factual tiers are used, because the published file also carries roughly 1,300 *mixed* domains whose inclusion would label every share of a mainstream outlet as misleading. An attached link preview is required, and the link must be to an article rather than a home page. Two of the 400 carry a link preview and no text of their own, so a study of what these posts say has 398 of them to work with while a study of how they spread has all 400.

One limitation survives all three filters and we state it wherever the sample is used: **a debunking links to what it debunks**, so an account refuting an unreliable article is labelled like one relaying it.

3.6 Transfer, adaptation and controls

Transfer is measured by applying the GossipCop-trained detector unchanged. Two variants are run: with account features, and with features replaced by a constant so that only topology survives the move. The second exists because account features are computed from different counters on each platform, and their failure alone could be a units artefact.

Adaptation is measured at three target-data budgets, 40, 100 and 200 cascades, comparing fine-tuning from the benchmark model against training from scratch on the same cascades with the

same seeds. A third route, replacing absolute sizes with dimensionless ratios, is also tested.

Controls cross two dimensions. Splits are made either cascade by cascade or by holding out entire source domains, the latter removing both account memorisation and the easier question. Cascade sizes are either left as they are or matched between classes. A one-parameter rule, *call it reliable above n accounts*, is fitted on the training half and reported beside the model in every condition.

4 Results

4.1 Detection on the benchmark

Table 1 reports macro-F1 on the test split, as the mean of three seeds with the seed-to-seed standard deviation.

Dataset	Features	GCN	GAT	Bi-GCN
PolitiFact	profile	0.772 $\pm$ 0.004	0.771 $\pm$ 0.004	0.776 $\pm$ 0.007
PolitiFact	bert	0.831 $\pm$ 0.002	0.831 $\pm$ 0.005	0.832 $\pm$ 0.004
GossipCop	profile	0.877 $\pm$ 0.007	0.874 $\pm$ 0.009	0.901 $\pm$ 0.009
GossipCop	bert	0.808 $\pm$ 0.015	0.881 $\pm$ 0.031	0.920 $\pm$ 0.024

Table 1. Propagation detection on UPFD, three architectures against two node feature sets.

Two readings follow from Table 1. Feature richness is the main lever, worth about +0.06 on PolitiFact. Architecture is the smaller one, and on the smaller corpus it is not a lever at all. The bidirectional model leads the better of the two baselines by +0.004 on PolitiFact *profile* and +0.001 on PolitiFact *bert*, against seed deviations of 0.002 to 0.007. The two GossipCop margins are larger, +0.024 and +0.039, against deviations reaching 0.031 on the second row, and they are not tested pairwise here. The comparison the next section tests is the one against a model that reads no edge.

4.2 Edgeless baselines

Table 1 says how well the detector scores. It does not say that the propagation is what it scored on, and the two are easy to confuse. The test is to hand the same node features to a model that cannot see a single edge.

Three such models are fitted, a logistic regression on the mean of the account features, one on the root feature alone, and one on the number of accounts as a single integer. None reads an edge. Table 2 reports them, and its last two columns give the graph model’s advantage over the strongest edgeless baseline of its row with a 95 % interval on that difference, from a paired bootstrap over 2,000 resamples of the test split repeated across three seeds. Every macro-F1 printed there is a mean over the three seeds on the split as drawn, as in Table 1, so the bootstrap enters the last two columns only. Bold marks the two rows whose interval excludes zero, one in each direction.

Dataset	Features	Accounts	Root	Size	Bi-GCN	Gap	95 % interval
PolitiFact	profile	0.790	0.767	0.415	0.775	-0.015	$[-0.064, +0.036]$
PolitiFact	bert	0.783	0.805	0.415	0.832	+0.027	$[-0.009, +0.064]$
GossipCop	profile	0.794	0.720	0.723	0.897	+0.103	$[+0.073, +0.129]$
GossipCop	bert	0.940	0.735	0.723	0.919	-0.021	$[-0.046, -0.004]$

Table 2. Edgeless baselines against the graph model, with the difference tested.

Four rows, four paired tests, and two of them separate in opposite directions.

On GossipCop with *profile* features the graph model leads by **+0.103** with an interval of $[+0.073, +0.129]$. That is the one place in this article where reading the propagation demonstrably beats ignoring it, and it is the configuration with the poorest node features, ten counters per account.

On GossipCop with *bert* features it is the other way. Averaging the account vectors and fitting a logistic regression reaches **0.940** against the detector’s 0.919, a gap of -0.021 with an interval of $[-0.046, -0.004]$. On the configuration carrying the highest figure in this article, the propagation structure supplies nothing a mean over the same features did not already hold.

The two PolitiFact rows establish nothing in either direction. Their intervals, $[-0.064, +0.036]$ and $[-0.009, +0.064]$, both contain zero on 221 test graphs. A structural result a mean-pool can match is not a structural result. On this benchmark that verdict falls on one of four configurations, its opposite on another, and the remaining two are silent.

The size column carries a cruder warning, and it splits the two corpora. On GossipCop a single integer, the number of accounts in the cascade, reaches 0.723. That is the same shortcut Section 4.9 finds in our own collected sample. On PolitiFact the same rule reaches 0.415, below chance, a threshold fitted on 62 training graphs that generalises to nothing. Cascade size is therefore a substantial shortcut on GossipCop and on our collected data, where the same family of rule reaches 0.902, and no shortcut at all on PolitiFact. Section 6.1 takes up what that contrast implies.

What the root vector holds under *profile*. The root of a UPFD cascade is the news item, so a column reporting 0.767 for “the root alone” under a feature set of ten account counters invites the question of whose account those counters describe. They are not constant: 310 of PolitiFact’s 314 root vectors are distinct, and 262 of a sample of 300 are byte-identical to an account vector appearing elsewhere in the corpus. The root therefore carries the statistics of the account that published the story, and those separate by class, the geolocation flag averaging 0.198 on the misleading side against 0.076 on the reliable one. This is not label leakage, but it is source identity, and it is the benchmark equivalent of the outlet-level confound we report against our own sample in Section 4.9. A model reading it is learning which publishers appear on which side.

The text inside those features. Under *bert* the root node carries an embedding of the news article, so a further question is whether the strong *bert* figures are the article being read through the root rather than a cascade being read at all. Masking the root’s feature vector, which leaves every edge in place and removes only the article, costs PolitiFact *bert* 0.832 ± 0.004 against 0.807 ± 0.003 , and deleting the root outright gives 0.801 ± 0.006 . The article is worth two to two and a half points, the sweep of Table 3 placing the masked model at 0.811 under its own fit. What remains after masking sits above the edgeless account baseline of 0.783 in Table 2, but that baseline’s own bootstrap interval, which the table does not print, runs to 0.837 and contains it, so this configuration establishes no more than the row above. Each of these figures is compared against the three-seed mean of the unablated model, 0.832, as reported in Table 1.

The cascade may be contributing here, and nothing in this configuration can show that it does.

4.3 Detection under cascade truncation

Table 3 reports macro-F1 with every model trained on complete cascades and evaluated on truncated ones, as means over three seeds with deviations of 0.002 to 0.014. Those deviations belong to this fit and are not the ones of Table 1, which reach 0.024 on the GossipCop *bert* row. *No edges* is a logistic regression on the mean of the account features, which reads no edge at any point, and *root masked* zeroes the article embedding in training and evaluation alike while leaving every edge in place, which is why it exists only for the two *bert* rows: *profile* roots carry no article. Bold marks the best model of each block. The sweep is a separate fit under the protocol of Section 3.2 and differences between rows are not tested in it, so Table 4 reports the paired test for the comparison it settles.

Corpus, features	Model	20%	40%	60%	80%	100%
PolitiFact, <i>profile</i>	Bi-GCN	0.787	0.792	0.787	0.784	0.775
	no edges	0.782	0.787	0.769	0.773	0.790
PolitiFact, <i>bert</i>	Bi-GCN	0.832	0.829	0.832	0.831	0.832
	Bi-GCN, root masked	0.814	0.809	0.818	0.815	0.811
	no edges	0.796	0.810	0.805	0.787	0.783
GossipCop, <i>profile</i>	no edges	0.599	0.733	0.775	0.794	0.794
GossipCop, <i>bert</i>	Bi-GCN	0.900	0.932	0.933	0.928	0.919
	Bi-GCN, root masked	0.922	0.954	0.961	0.963	0.963
	no edges	0.885	0.935	0.945	0.945	0.939

Table 3. Detection against the share of the cascade observed.

How much of a cascade the detector needs is settled by truncating it, and Table 3 reports the result. On PolitiFact with text-rich features the curve is flat to within 0.003 across the five levels. At scale, on 3,826 test cascades, it is not flat but shallow: **twenty per cent of a cascade yields 0.900** against a peak of 0.933 and 0.919 for the complete one, and two to four seed deviations separate that cell from the rest of its row. A fifth of the cascade therefore buys most of the available accuracy and not all of it.

Read at face value that is the most useful result in this article. Two alternative explanations for it are tested below, and the second holds on one of the two corpora.

First alternative, on PolitiFact: the root the truncation never removes. Breadth-first truncation always retains node 0, node 0 carries the article embedding under *bert*, and a model leaning on that one vector would score identically at every level because nothing it uses was ever taken away. Rerunning the sweep with the root’s features masked, which removes the article and leaves every edge, keeps the curve flat: in Table 3, 0.814 at 20 % against 0.811 at 100 %, and the row spans 0.009 where the unablated model spans 0.003. The whole curve sits about two points lower, uniformly, which is the article’s contribution. On this corpus the explanation is ruled out. On GossipCop the same operation does something else entirely, and Table 4 reports it.

Second alternative, and it is not ruled out. The detector mean-pools its node representations before classifying. The mean of a feature over the first fifth of the nodes estimates the mean over all of them, and it converges quickly. A model whose decision is largely a function of that mean would hold its score under truncation for a reason with nothing to do with propagation.

Masking the root does not test this, and neither does anything else reported above.

The test removes the graph itself. Logistic regression is fitted on the mean of the account features of complete training cascades and evaluated on test cascades truncated to the same fractions, reading no edge at any point.

On GossipCop *bert* the edgeless curve reaches 0.885 from a fifth of the cascade and 0.939 from all of it. Most of the early-detection behaviour is therefore available with no graph at all, and a large part of “one fifth is enough” is a statement about how quickly a sample mean converges.

The two curves do not sit on top of each other, and the place they part is where any remaining structural claim has to live: 20 %, where the graph model reads 0.900 against an edgeless 0.885. Placing a seed deviation beside a split interval does not test that difference. The two quantify different sources of variation, one across seeds at a fixed split and one across splits at a fixed fit, and the graph model carries both while a table can report only one. The difference itself has to be measured.

The test is the one used on the content side of this project. Both models are scored on the same bootstrap resample of the 3,826 test cascades, the difference is recorded, and the procedure is repeated for each seed of the graph model so that both sources of variation enter the same interval.

Table 4 tests that comparison with a paired bootstrap over 2,000 resamples of the test split, repeated across three seeds of the graph model and pooled. As in Table 2, the bootstrap enters the last two columns only: the macro-F1 are means over the three seeds on the split as drawn, and are the cells of Table 3 at 20 %. The interval is on the difference, not on either score.

Model at 20 %	Macro-F1	Gap vs edgeless	95 % interval
Logistic regression, no edges	0.885	-	-
Bi-GCN	0.900	+0.015	$[-0.005, +0.038]$
Bi-GCN, root masked	0.922	+0.037	$[+0.023, +0.051]$

Table 4. The graph model against a model with no edges, on GossipCop *bert* at 20 % of the cascade.

Table 4 gives two answers and the second is the one that matters. **The detector as configured does not separate:** its interval against the edgeless baseline runs from -0.005 to $+0.038$ and contains zero. Read on its own, that would leave the graph indistinguishable from a model that cannot see an edge at the level where the two curves are furthest apart.

Masking the root reverses it. Zeroing the article embedding raises the graph model to 0.922, above its own unmasked score, and the gap against the edgeless baseline becomes $+0.037$ with an interval of $[+0.023, +0.051]$ that excludes zero on every resample. The same operation at the other end of the sweep, on complete cascades, raises the detector to 0.963 and turns its -0.021 deficit into a lead of $+0.024$, each measured against the edgeless baseline of its own table, 0.940 in Table 2 and 0.939 in Table 3, with an interval of $[+0.016, +0.031]$. The article embedding was not helping on this corpus, and its removal exposes a structural signal underneath it at both levels we tested pairwise. The three intermediate levels are not tested, and their gaps against the edgeless baseline, $+0.019$, $+0.016$ and $+0.018$, are the three narrowest of the sweep.

A dilution account would explain this: at 20 % a GossipCop cascade holds a handful of accounts, so the root’s 768 dimensions dominate a mean pool taken over so few, and the effect should fade as cascades fill out. The measurement runs against it on both corpora. On GossipCop the gain is $+0.022$ at 20 % and grows to **$+0.044$** at 100 %. On PolitiFact it is negative at all five levels, between -0.015 and -0.021 : there the article embedding *helps*.

What the article embedding is worth to a propagation detector depends on the corpus, and it changes sign. On GossipCop it costs between 0.022 and 0.044 macro-F1 and the cost grows with the cascade. On PolitiFact it earns between 0.015 and 0.021, steadily. Ten measurements, two corpora, and no crossing, drawn in Figure 2. No single statement about what this feature contributes holds across the benchmark, which makes it a property of the resource rather than of the model we trained on it.

Figure 2 runs the same ablation across the same five truncation levels on both subsets of the benchmark. On GossipCop the masked curve sits above the unmasked one at every level, so removing the article *helps* the detector, and the gap widens as the cascade fills. On PolitiFact the masked curve sits below the unmasked one at every level, so there the article helps instead. Ten measurements, two corpora, no crossing, and the cells are those of Table 3.

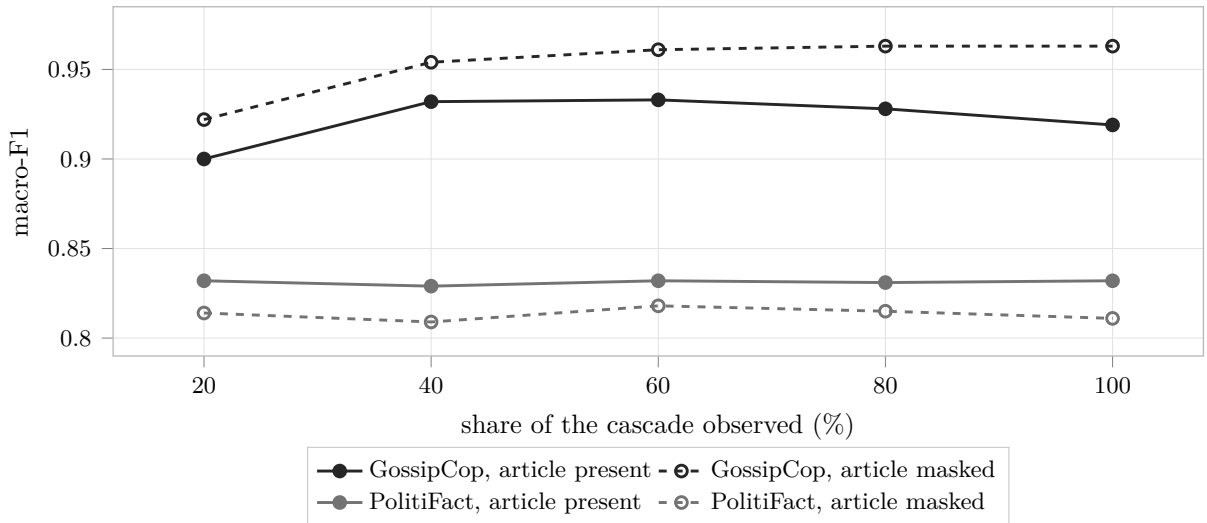


Figure 2. The article embedding changes sign between corpora.

Two qualifications bound the section. Averaging does not automatically flatten a curve: on GossipCop *profile* the edgeless sweep of Table 3 climbs from 0.599 to 0.794, a spread of 0.196, so a flat curve is not something a mean produces by itself. The PolitiFact rows are not evidence in either direction. Their gap intervals in Table 2 span 0.100 and 0.073 on 221 test graphs, and that paired test finds nothing there on complete cascades.

What survives the two qualifications is that a verdict is available from a fifth of a cascade, and that on GossipCop *bert* it is better than a mean-pool once the article is masked out of the root, while on the two PolitiFact rows the two cannot be separated. Section 6.2 reads that against the study the architecture comes from.

4.4 Cascade shape by class

Table 5 averages over all cascades of each corpus, 314 on PolitiFact and 5,464 on GossipCop. *Accounts* counts the nodes of the cascade, *depth* the longest chain of reshares from the source, *breadth* the largest number of accounts at any one hop, and *direct shares* those that took the story from the source itself.

Dataset	Class	Accounts	Depth	Breadth	Direct shares
PolitiFact	fake	114.9	3.62	71.8	62.7
PolitiFact	real	146.6	3.89	87.7	79.4
GossipCop	fake	76.2	2.48	62.2	60.6
GossipCop	real	38.9	2.54	27.9	26.1

Table 5. Average cascade shape by class.

Table 5 gives the averages and Figure 3 the ratios between the classes. On GossipCop, false stories reach 1.96 times as many accounts, are 2.23 times broader and are shared **2.32 times more often directly from the source**, while travelling no deeper: depth is 2.48 against 2.54. They are **amplified rather than relayed**, so what separates the classes is concentrated near the source. That is consistent with a verdict being available early, though the edgeless sweep of Table 3 shows it is not sufficient to explain it: a mean over the same features is available just as early.

Figure 3 plots how much larger a fake cascade is than a real one, by measure and by corpus. Above the dashed line at 1.0 the false stories are the larger, below it the true ones are. What generalises is that the classes differ in shape.

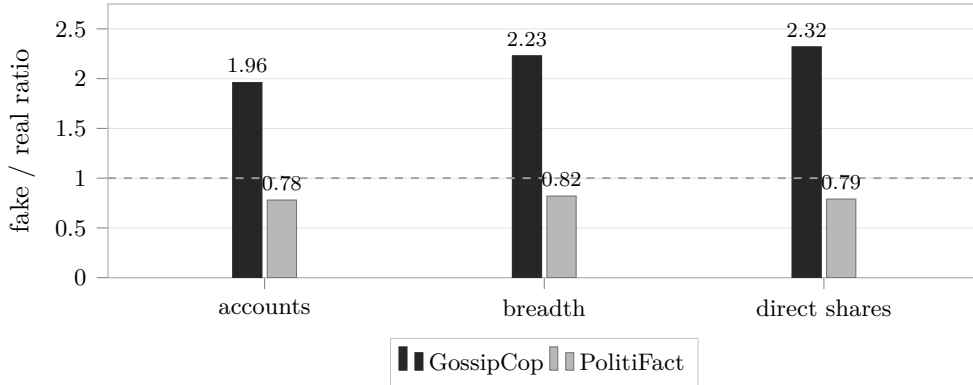


Figure 3. The direction reverses between corpora.

The direction is not universal, which constrains what may be claimed from it. On PolitiFact the ordering reverses: real cascades are the larger ones. What generalises is that the two classes have *different* shapes, which is what the detector exploits, and Section 6.3 takes up which of the two readings travels.

4.5 Influence ranking against veracity

Merged on account identity, the 400 labelled collected cascades form a genuine network: 39,188 accounts and 63,438 edges, 8,629 accounts appearing in more than one cascade, and a maximum k-core of **12**. The 400 cascades carry 69,337 interactions between them, and the merged graph is simple, so the 5,899 that disappear are pairs of accounts that interact in more than one cascade. On the benchmark, cascades are disjoint trees with no shared accounts, so k-core is constant at 1 and the measure is vacuous. Real data is what makes the ranking meaningful at all.

The evaluation then requires care with the baseline, and this is where the result lives. The two classes hold 200 cascades each, so at cascade level the sample is balanced. Reliable cascades, however, average **259** accounts against **44** for misleading ones. Counting account *slots* rather

than cascades, the misleading side holds only **14.5 %** of the total, and that, not 50 %, is what chance looks like.

Table 6 reports whether structural influence identifies the accounts carrying misleading material, against a baseline set by unequal cascade sizes rather than by the class balance.

	Share on the misleading side
Chance baseline	14.5 %
Accounts in more than one cascade	13.9 %
Top 100 by influence	13.4 %

Table 6. Influence against veracity.

Table 6 gives the answer, and the top hundred sit *below* chance. Fifty-three of them appear only in reliable cascades and two only in misleading ones. A rank correlation of +0.086 does survive a stratified permutation test that shuffles influence among accounts of equal activity ($p = 0.012$), but it explains under 1 % of the variance.

Cascade size is the confound this article keeps finding, so it is worth closing here too. An account that lands in a large cascade collects more edges in the merged network and earns a higher reach and PageRank for reasons unconnected to its behaviour, and large cascades on this sample are the reliable ones. Holding activity *and* the size of the cascades an account joined fixed, the association survives at $p = 0.032$. It survives because the route is narrow to begin with: influence and mean cascade size correlate at only +0.018 ($p = 0.09$), while size and misleading share correlate at -0.253 .

The null itself is not centred on zero, producing a mean rank correlation of +0.073 against the +0.086 observed, and the obvious explanation for that is not the correct one. It is not that the strata are degenerate: only 33 of the 8,629 accounts sit in a stratum of one, where a shuffle cannot move them. It is that shuffling inside strata leaves the between-strata component of the correlation untouched, so a global rank correlation is being compared against a null that only tests the within-stratum effect. The difference between the two is not an estimator of anything.

Measuring the within-stratum effect directly, by centring each stratum on its own mean rank before correlating, gives $\rho = +0.021$ across the 8,629 accounts. That is what influence contributes once activity and cascade size are held fixed: detectable on this many accounts, and negligible.

Structural centrality locates who drives diffusion without identifying who diffuses misinformation. On this platform the most central accounts are newsrooms and well-followed commentators, and they predominantly carry credible material. Section 6.4 sets that against what the superspreader literature reports.

4.6 Composition of the collected cascades

Before the transfer is measured, the two sides of the comparison must be described, because they are not made of the same material. Figure 4 draws four of them.

In Figure 4 every dot is an account and every line is one account taking the story from another, and the orange dot is the source. The layout is force-directed, so position on the page carries no measurement: what it shows is connectivity, not distance from the source. The top row links to outlets the Iffy Index rates low for factual reporting, the bottom row to outlets rated high. A solid line is a repost and a faint one a reply, and the composition this section is about is visible in the difference: *rt.com* is 70 % reposts and reads as solid spokes, while the other three run from 72 to 86 % replies and trail off into faint chains. Three of the four are a single dense core of accounts taking the story straight from the source, with thin threads leading away from it.

apnews.com is the exception and the interesting one, since it has a second core, an account that took the story from the source and was then itself relayed by 157 others, against 272 for the source. On the other three the largest secondary relay reaches nine. Each panel is fitted to its own frame, so the panels compare shape and not size.

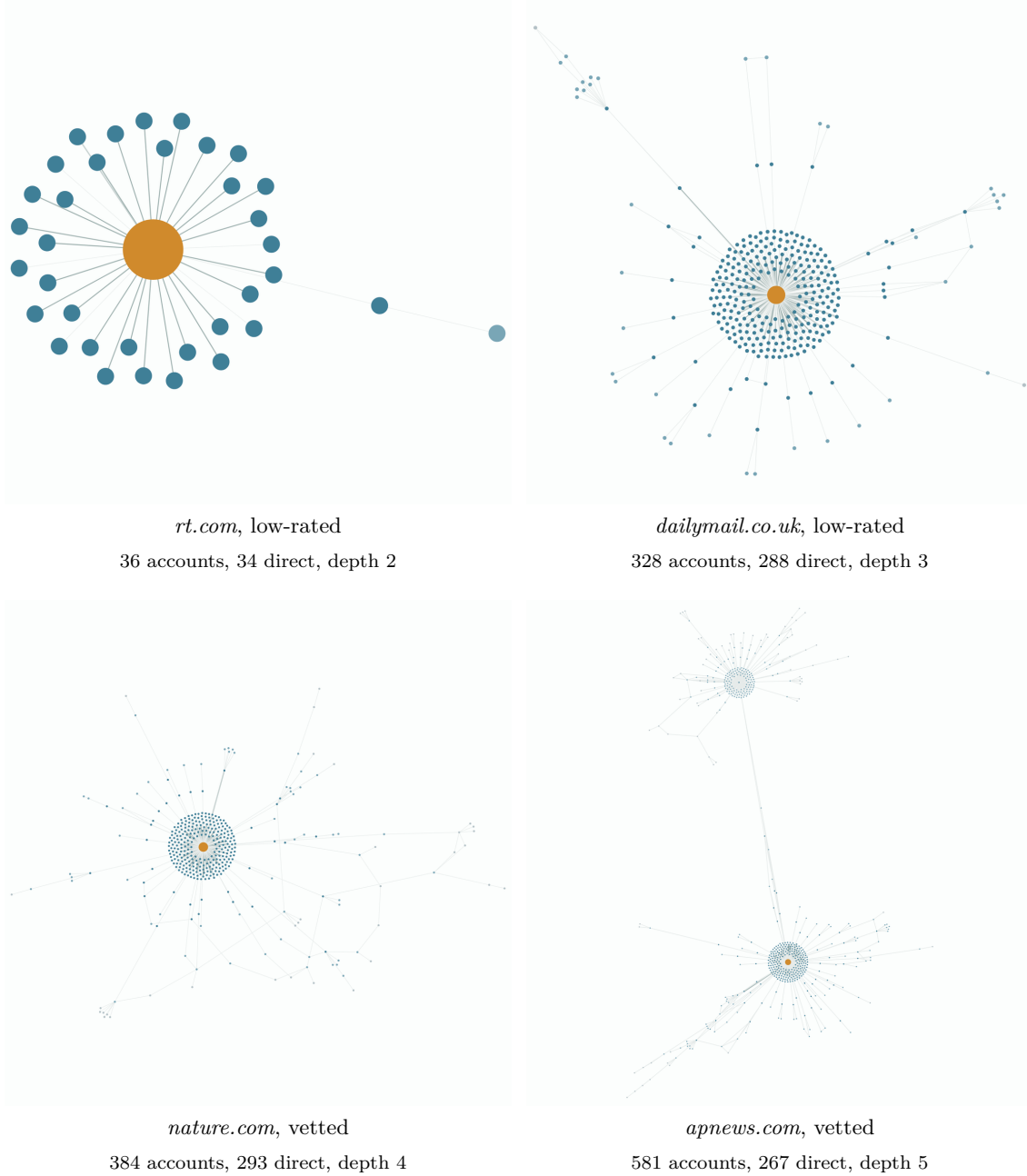


Figure 4. Four collected cascades, drawn.

A UPFD cascade is a retweet tree. A collected cascade is not. **Two thirds of its edges, 46,066 against 23,271, are replies rather than reposts.** The two acts are not equivalent. A repost carries the post to an account's own followers, a reply travels nowhere, and under a doubtful claim a reply is frequently a correction. Counting both as diffusion credits pushback as spread.

The reposts are also flatter than they appear. The collector asks which accounts reposted the *original* post and never repeats the question for the reposts themselves, so across all 400 cascades

every repost edge leaves the root. Whatever depth these objects have comes from the reply tree, and a reported depth of five is five levels of conversation, not five hops of diffusion.

Neither point invalidates what follows, which is measured on these objects as they are. Both bear on how it should be read, and both supply a mechanism for the next section: a detector trained on retweet trees is being asked about objects a third of whose edges are retweets.

4.7 Cross-platform transfer

Table 7 applies it to the collected cascades with no fine-tuning of any kind, reporting macro-F1 as the mean of three seeds with the seed-to-seed standard deviation. *Strict subset* restricts the collected side to the cascades whose linked outlet the Iffy Index rates in its lowest factual tier, which is the cleanest half of the labels. A constant answer on a balanced binary task scores 0.333, a floor the strict subset need not meet since it restricts only the misleading side. Bold marks the Bluesky column, which is the result the table exists to report.

Variant	In domain	Bluesky	Strict subset
With account features	0.901 $\pm$ 0.009	0.378 $\pm$ 0.032	0.456 $\pm$ 0.009
Structure only	0.754 $\pm$ 0.040	0.303 $\pm$ 0.022	0.330 $\pm$ 0.014

Table 7. Cross-platform transfer, the GossipCop-trained detector applied unchanged.

A constant answer on a balanced binary task scores 0.333 in macro-F1. The figures of Table 7 sit at 0.378 and 0.303, one just above that floor and one below it, and the predictions do pile onto a single class. This is not degradation, it is collapse.

The two variants separate the explanations. Account features are computed from different counters on each platform, so their failure could have been a scaling artefact. The structure-only variant, in which nothing but topology survives the move, collapses as well. What the model learned about the shape of Twitter cascades therefore does not describe Bluesky cascades.

That objection deserves more than one variant, because the mismatch it points at is severe. Three of the ten counters have no equivalent on the target platform and the collector writes them as zero: no verification flag, no geolocation, no list membership. Of the seven that remain, four sit orders of magnitude from their benchmark counterparts, the follower count by a factor of 14,000 and the favourite count by 4,600, because the collector’s compression was written to mirror a normalisation that had not been verified. A model handed inputs that far outside its training marginals would collapse whatever the domain gap was. Mapping each surviving counter onto the benchmark’s own marginal for that counter, a rank-preserving transformation that moves the scale and leaves the ordering of accounts untouched, lifts the transfer to 0.443 ± 0.029 , against the 0.378 ± 0.032 of Table 7. The repair is worth 0.065 and still lands below chance, so that alignment does not account for the gap. Three of the ten counters have no counterpart on the target platform and are written as zero, which this repair does not address. What the collapse costs, and whether it can be repaired, is the subject of the next two sections, and Section 6.5 reads it against the resources this area is built on.

4.8 Cost of adaptation

Table 8 measures adaptation to the target platform on the same cascades and the same three seeds at every budget, with 60 epochs in every arm. The last column takes the best of learning rates 0.01, 0.005 and 0.002 for the fine-tuned model, and bold marks the leading arm at each budget.

Cascades used	From scratch	Fine-tuned, matched	Fine-tuned, best rate
40	0.722 $\pm$ 0.007	0.582 $\pm$ 0.099	0.583 $\pm$ 0.101
100	0.765 $\pm$ 0.032	0.651 $\pm$ 0.012	0.651 $\pm$ 0.012
200	0.777 $\pm$ 0.021	0.714 $\pm$ 0.018	0.716 $\pm$ 0.018

Table 8. The cost of recovery.

Three questions, in the order that makes them answerable.

Whether a signal is present at all. Trained and tested on the collected cascades alone, the same architecture reaches macro-F1 0.767 ± 0.023 . Yes, so the collapse was a failure of transfer rather than of what is being predicted.

The cost of recovery. Table 8 gives it, and the answer depends on a protocol choice that is easy to get wrong. Fine-tuned at 30 epochs and a learning rate of 0.002 against a from-scratch arm at 60 and 0.01, the benchmark-initialised model reads 0.425 at forty cascades and looks actively obstructive. Given the same epochs and rate it reads 0.582, and given the best of three rates 0.583. Half the apparent interference from the source domain is a difference in optimisation budget, and the comparison is only informative when the two arms are matched.

Matched, training from scratch still leads at every budget, by 0.140 at forty cascades, 0.114 at a hundred and 0.063 at two hundred, so a benchmark initialisation is not the head start it is usually assumed to be. Both arms improve monotonically and the lead more than halves as target labels accumulate, so the gap does close as target data is added. Three points do not support extrapolating the trend beyond them.

One number in the table should be read with suspicion. The fine-tuned arm at forty cascades carries a standard deviation of 0.099 across three seeds, an order of magnitude above every other cell. Forty cascades is one gradient batch, and starting from benchmark weights makes the outcome depend heavily on which sixty steps are taken. The mean of 0.582 is the least trustworthy figure here, and the direction of the result does not rest on it.

Avoiding new labels. Replacing absolute sizes with dimensionless ratios yields 0.802 in domain, so the features are perfectly adequate descriptors. Transferred, they give **0.215 $\pm$ 0.009**, worse than the raw features they were meant to improve on. Scale invariance was the wrong hypothesis. Across the three questions, then, the signal is present on the target platform, the cheapest route to it is a few dozen fresh labels rather than a ported model, and scale invariance is not a substitute for either. Section 6.5 draws the recommendation that follows.

4.9 Size-matched control

Table 9 reports macro-F1 as the mean of three seeds with the seed-to-seed standard deviation, against a chance level of 0.500. Size matching leaves 39 pairs, so the two matched rows draw their test halves from 78 cascades rather than 400. The test halves hold 160, 116, 32 and 47 cascades from the first row down, the domain-grouped splits holding out whole outlets and therefore landing on uneven sizes. The 0.111 on the second row means the size rule is unstable under a domain split, not that it is reliably strong there. The 0.309 on the last row is the tightest cell in the table and sits far below chance, reproducibly: holding out whole outlets inverts the relation between size and label. The inversion is not accounted for here.

Condition	One size threshold	Bi-GCN
Split by cascade, all sizes	0.902 ± 0.021	0.768 ± 0.040
Split by domain, all sizes	0.912 ± 0.111	0.713 ± 0.068
Split by cascade, size matched	0.504 ± 0.025	0.685 ± 0.045
Split by domain, size matched	0.309 ± 0.002	0.517 ± 0.055

Table 9. What the collected labels are predictable from.

The 0.767 of the previous section, which the cascade-level row of Table 9 reproduces at 0.768, invites a structural reading. That reading is tested here before it is made. Reuters, AP and NPR carry 260 to 340 accounts per cascade, and Gateway Pundit, Newsmax and RT carry 36 to 62. That is the audience of the outlets we sampled, not a property of misinformation.

Table 9 asks what the labels are predictable from. They are largely predictable from size alone: a rule that calls a cascade reliable above n accounts reaches **0.902** where the detector reaches 0.768, and it survives the harder domain-grouped split, though with a seed deviation of 0.111 that says more about instability than about strength. Matching the sizes away drops the rule to chance and leaves the detector at **0.685**.

A structural signal therefore survives on this sample under the cascade-level split, and it is **much smaller than the uncontrolled 0.767 suggested**. It does not survive the harder of the two splits. Under the domain-grouped one, the protocol the companion paper recommends for reuse of this sample, with sizes matched, the detector reads 0.517 ± 0.055 and its seed deviation straddles chance. Section 6.6 places both controls against the studies they come from, and states what is new here.

5 An external content signal as a node feature

The detector described so far reads structure and nothing else. It is one of two sub-modules of a larger system, the other being a content classifier described in a companion paper. This section reports what that external signal adds to this model, from this side of the interface.

Figure 5 applies the system of this section to a collected cascade of 581 accounts. The content model reads the wording and answers *misleading*, 0.353, while this model reads the cascade and answers *reliable*, 0.603. Writing the content score onto the root and letting this model decide again gives 0.628, so the score moved the structural verdict by +0.026 and moved it *away* from what the content model itself concluded. That is the behaviour a weak interface is designed to permit: the score is one more number per account and not a second opinion to be averaged in, and this model is free to use it as evidence about the cascade rather than as a verdict about the story. One cascade is an illustration and not a measurement, and Table 10 carries the measurement.



Figure 5. One cascade, read by both halves of the system.

5.1 Design of the coupling

Recent surveys place fused content-plus-structure architectures at the centre of the field [6, 11]. We depart from that on purpose. The content model’s credibility score enters this model as **one additional feature column on the root node of the cascade**, and nothing else. No representation is shared and no loss is joint.

The reason is methodological. A weak, explicit interface makes the contribution of the external signal measurable by **ablation**: the model can be run with the column, with a column of random values, and without it at all. Three numbers and one attributable difference. A fused model would very likely score higher and would leave us unable to say what the text contributed, which for a study whose subject is what these signals are actually worth would defeat the purpose.

The score is produced by a small multilayer perceptron over the news article embedding this benchmark supplies at the root. Fitted on the published training split and measured on the held-out one, it reads 0.823 ± 0.004 on PolitiFact and 0.735 ± 0.000 on GossipCop on its own, the first of those sitting 0.018 above the logistic regression the companion paper fits on the same input. It is not the short-claim classifier the companion paper reports at 0.63, which was trained on another corpus at another unit of text. Scores are cross-fitted over five folds so that no story is scored by a model that saw it, and a random-score control accompanies every condition.

5.2 Contribution of the content score

Table 10 ablates inside the propagation model, as the mean of three seeds with a per-cell standard deviation between 0.004 and 0.017. The +0.056 of the first row therefore clears its own noise and the +0.003 of the second does not, and bold marks that gain as the only one in the table larger than its own deviation. The *graph only* column runs 0.004 to 0.011 from the Bi-GCN of Table 1, being the same architecture on the same splits fitted on an input widened by one always-zero column. On PolitiFact *profile* that difference of 0.011 exceeds the seed deviation of either table.

Dataset	Node features	Graph only	+ random	+ content
PolitiFact	profile	0.787	0.783	0.843
PolitiFact	bert	0.828	0.828	0.831
GossipCop	profile	0.909	0.909	0.905
GossipCop	bert	0.914	0.935	0.929

Table 10. What an external content score adds.

Read from this side, Table 10 says something about *our* model rather than about the content classifier.

On PolitiFact, where nodes carry only the ten account counters, the structural model is **not saturated**: an external text signal recovers a quarter of the remaining error, from 0.787 to **0.843**, and the random control moves nothing. Whatever the graph was missing, part of it was linguistic, and the architecture had no way to reach it.

That row is also where the two halves of this article meet. Table 2 puts an edgeless logistic regression at 0.790 on this configuration, which the same table declares indistinguishable from the propagation model that reads every edge. Adding one column of text moves the graph model to the 0.843 of Table 10, past both. The structure was not what made the difference on this corpus, and it was not inert either. It was waiting for a signal it could not compute itself.

Where nodes already embed each account’s posting history, the same signal adds +0.003. The 768-dimensional features were already carrying the linguistic information, by a longer route.

The fourth row disciplines the rest. There the *random* control gains +0.021 against the genuine score’s +0.015 on the same row, and the genuine score gains −0.004 on the row above and +0.003 on PolitiFact *bert*. A control that outperforms the signal it controls for means the block is noisy, and we draw nothing from it.

6 Discussion

The sections above are measurements. This one reads them, against the studies whose methods and tests they reuse and against what a system built on them would have to assume. Table 11 sets out the first of those comparisons in one place: each study, the method or test taken from it, and what that yields on our data, with our own result on the last row.

Study	Method or test reused	Outcome here
Kipf and Welling [9]	Graph convolution over the cascade	0.772 PolitiFact, 0.877 GossipCop, <i>profile</i>
Veličković et al. [10]	Attention-weighted convolution	0.771 and 0.874 on the same two
Bian et al. [4]	Two directed views of the cascade, pooled	0.776 and 0.901, the architecture we adopt
Dou et al. [5]	The benchmark, its node features and its edgeless baselines	The edgeless baseline reaches 0.940 on GossipCop <i>bert</i> , above the graph model
Kitsak et al. [13]	k-core as a predictor of spreading, inside our composite score	Ranking well defined only on real data, and 13.4% against a 14.5% baseline

Study	Method or test reused	Outcome here
Pelrine et al. [7]	A simple predictor run as a control on a reported result	Cascade size alone reaches 0.902 where the detector reaches 0.768
Wu and Hooi [8]	Event-separated evaluation against spurious cues	Holding outlets out inverts the size relation, 0.309
This article	Bi-GCN with the root article masked, tested against the edgeless baseline under a paired bootstrap	0.963 on GossipCop, +0.024 over the edgeless baseline, interval [+0.016, +0.031]

Table 11. Prior work whose method or test this article reuses.

6.1 What a benchmark figure attributes

The three architectures of the first rows separate from each other by 0.005 to 0.024 on these corpora. The edgeless baseline of the fourth row moves the same comparison by as much as 0.103 in one direction and 0.021 in the other. A margin between architectures is therefore smaller than the margin between reading the graph and not reading it at all, and only the first of the two is reported by the studies the second is usually compared against.

Two of our measurements say where that second margin comes from. The first is cascade size, and whether a benchmark can be gamed by counting is a property of the benchmark rather than of the task: one integer reaches 0.723 on GossipCop and 0.415 on PolitiFact, so the resource the field treats as standard is gameable on one of its two halves and immune on the other. The second is the news article the benchmark places at the cascade root. Masking it is worth ± 0.02 to ± 0.04 depending on the subset, which exceeds the architectural differences this literature reports, and it is a feature inherited by every study using this resource and ablated by none of them.

Why that contribution changes sign between the two subsets is not settled here. The obvious candidate is the size of the training set, since PolitiFact supplies 62 cascades and GossipCop 1,092. With 62 examples a dense, informative feature at a fixed position is worth having, and with 1,092 there is enough structure to learn that the same feature is something to look past. We offer that as the reading the measurement suggests and not as a result, because two corpora that differ in size also differ in domain, in class balance and in cascade shape, and we cannot separate those here.

6.2 Early detection and what carries it

A verdict is available while a story is still spreading, which is the condition any deployment has to meet. On GossipCop *bert* the graph model delivers it better than a mean-pool once the article is masked out of the root, and on the two PolitiFact rows, where we cannot show that, a mean-pool is the cheaper thing to build. The early-detection behaviour Bi-GCN was introduced with [4] is therefore reproduced here at a scale the original study did not cover, and qualified by a baseline it did not run.

The qualification carries more weight than the reproduction. A large part of the claim that one fifth of a cascade is enough turns out to be a statement about how quickly a sample mean converges, since a model reading no edge at all holds its score under the same truncation. Only the masked configuration separates from that statement, and it does so at both ends of the sweep.

6.3 Cascade shape and what travels between corpora

That falsehood produces the larger cascade is a property of GossipCop’s celebrity-news domain rather than of misinformation. The finding that false news travels farther [1] is reproduced here on one subset and reversed on the other, which is a caution about generalising from a single corpus rather than a contradiction of that work. What survives the move between the two is the weaker statement, that the classes differ in shape, and it is the one the detector exploits.

6.4 Influence and veracity

That a small number of accounts drive a disproportionate share of low-credibility spread [14, 15] is not in question here. What our result denies is that structural centrality, of the kind k-core position measures [13], is a way of finding them. Our composite follows the strand running from Kempe et al. [12] through Kitsak et al., the ranking it produces is stable under reweighting, and the hundred most central accounts on our sample sit below the chance baseline for misleading material.

The distinction is operational rather than semantic. Centrality answers a question about diffusion, and the accounts it returns on this platform are newsrooms and well-followed commentators, so a moderation system built on it would act on the wrong set.

6.5 Crossing platforms

A system evaluated only in domain would have looked ready for deployment. Every standard resource in this area [17, 18, 5] comes from the same platform era, so a result of this kind cannot be obtained from the benchmarks alone, which is the practical form of the critique of single-corpus evaluation [23].

What the collapse costs is small in absolute terms and awkward for the usual remedy. Forty cascades labelled in an afternoon recover most of the available accuracy, and the benchmark initialisation contributes little without obstructing once both arms are given the same optimisation budget, half the apparent interference from the source domain having been a difference in that budget. The recommendation that follows is the opposite of the one transfer learning would suggest: **deploying propagation-based detection on a new platform should not begin by porting a model but by labelling a few dozen cascades on the target platform**, which outperformed every transfer configuration we tested at every budget we tried.

6.6 Shortcuts in a sample built to avoid them

Neither control we ran on our own data is novel. The domain-grouped condition is Wu and Hooi’s event-separated protocol [8] with outlets in the role of events, and a single trivial feature outscoring a trained model is the situation Pelrine et al. documented on Twitter16 [7]. What is new is the particular shortcut, cascade size induced by sampling outlets rather than stories, and the fact that a sample assembled to escape the benchmarks turns out to carry one of its own.

That is why both checks are reported here in full rather than folded into a limitations paragraph. Applied to the benchmark figures this literature reports, they would be informative, and they are not standard practice.

6.7 Consequences for the two-signal design

Three of our own results constrain how this interface should be understood.

An external content signal is worth acquiring where the node features cannot supply it. A classifier reading the article, at 0.823 on this corpus, closes a quarter of our remaining error on PolitiFact where our nodes are ten counters, and closes none on GossipCop under the same

feature set. For a system architect the consequence is that a content module should be budgeted against what the rest of the system can already see, on the corpus it will run on, and not against its own accuracy.

Four in seven real collected posts state no checkable claim at all, again per the companion paper, on a rule-based estimate wide enough that the proportion should be read as an order of magnitude and not a measurement. Where the text asserts nothing, the structural signal is the only available evidence rather than a complement, which is an empirical argument for this module and not an architectural preference.

And the gain is conditional on what this model can already see. A system integrating the two signals should therefore decide whether to pay for a content model by looking at its own node features.

Finally, a scope note. The ablation is computed on benchmark cascades throughout, and no content score derived from collected text enters this model anywhere in this work, so the link artefact the companion paper reports on that text does not touch these figures.

7 Conclusion and limitations

A propagation model reaches macro-F1 0.920 on the standard benchmark. A logistic regression over the same node features, with every edge deleted, reaches 0.940. Tested pairwise on all four configurations, the graph model leads on one by +0.103, trails on another by -0.021, and is indistinguishable on the two built from 221 test graphs. On GossipCop the size of a cascade alone, a single integer, reaches 0.723.

The configuration it trails on is one of the two where UPFD attaches the news article to the root, and on that corpus the loss disappears when that attachment is masked. Masking the article embedding is worth +0.022 to +0.044 across the GossipCop sweep, and with it masked the detector reaches 0.963 and leads the edgeless baseline by +0.024, interval [+0.016, +0.031]. On PolitiFact the same ablation costs 0.015 to 0.021 at every level. A benchmark feature inherited by every study using this resource, and ablated by none, is worth between -0.021 and +0.044 depending on the subset, which is larger than any architectural difference reported here. Detecting it requires a baseline that reads no edge, which this literature does not routinely construct.

A verdict is available from a fifth of a cascade, and an edgeless baseline delivers one too. At 20% the paired test cannot separate the two, and on complete cascades it separates them in the baseline's favour, -0.021 with an interval of [-0.046, -0.004]. Masking the article embedding out of the root reverses both readings: the graph model then leads by +0.037 at 20% and by +0.024 on complete cascades, intervals [+0.023, +0.051] and [+0.016, +0.031]. On GossipCop, with the article masked, the propagation structure carries something a mean over account features does not, at both levels we tested. On GossipCop the class shapes are consistent with this, misleading stories being amplified instead of relayed, so what separates them there is concentrated near the source.

Structural influence, measured where the measure becomes meaningful, does not identify the accounts carrying misleading material. Against a correctly constructed 14.5% baseline the hundred most influential sit at 13.4%. Measured inside strata of equal activity and cascade size, what influence adds is $\rho = +0.021$.

Carried to another platform the detector collapses, and the one repair we tested does not account for it. Aligning the account counters onto the benchmark's own marginals lifts the transfer by 0.065 and leaves it below chance. Three of the ten counters have no counterpart on the target platform and are written as zero, so other preprocessing differences remain untested. Starting from the benchmark still loses to training from scratch on a few dozen target cascades, by 0.14

at forty target cascades once both arms are given the same epochs and learning rate, and by 0.06 at two hundred. On that collected sample a one-parameter rule on cascade size outscores the detector, and once sizes are matched the structural signal that remains survives the cascade-level split and not the domain-grouped one, far below the uncontrolled figure. An external content signal, injected as a single feature column, recovers a quarter of the remaining error on PolitiFact with account counters at the nodes, and none on the three other configurations, which places the value of combining the two signals on the configuration rather than on the combination itself.

Limitations. The collected sample mixes reposts with replies and derives its depth from the reply tree alone, so its depth and virality figures are not comparable term by term with the benchmark’s. Its labels come from the rating of the linked outlet, so an account refuting an unreliable article is scored like one relaying it, and two retained domains publish ordinary reporting alongside a poor factual record. Three of the ten account features are constant throughout, the AT protocol exposing no verification flag, no geolocation and no list membership. The size-matched domain-grouped condition rests on 47 test cascades and should be read as indicative. UPFD carries no timestamps, so the early-detection protocol uses breadth-first order as a proxy for arrival order, and the curve should be read accordingly. Outside the ablations, which intervene on a feature and report what changes, every result here is an association and not a cause.

Data and code availability. An anonymised sample of the 400 collected cascades, with account identities pseudonymised and outlet links removed, and the code for every experiment reported are available from the authors on request, and will be released publicly on publication. The link removal is deliberate: Section 4.9 and its counterpart on the content side show what a classifier fitted on this sample reads when the link is left in. Each result is reproducible from a single command with fixed seeds.

Declaration of AI use. During the preparation of this work, the authors used Claude (Anthropic), a large language model, to assist with writing and debugging the code for the experiments and the data processing, and with drafting and editing the text of the manuscript. Every figure reported was produced by running the experiment code. The authors reviewed, verified and edited all output and take full responsibility for the content of this article.

8 References

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